
Beyond Data-Free Distillation: Serverless Knowledge Distillation for Heterogeneous Federated Learning

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Abstract

Federated Learning (FL) struggles with non-IID client data, where Federated Averaging (FedAvg) often fails due to conflicting local updates and skewed distributions, because parameter averaging can disrupt the functional consistency between local and global models. Most existing approaches restrict the local update on the client, ignoring the performance drop caused by direct aggregation of the global model. We study knowledge distillation as a way to rectify, or replace, parameter averaging: after each training round, knowledge is distilled from the client models into the global model in a multi-teacher, single-student setup. We first extend data-free, noise-engineered knowledge distillation to FL, so that aggregation requires no data at all. We find that in this regime, pure data-free distillation fails to provide competitive performance and converges slowly, underperforming even FedAvg. We therefore introduce a decentralized, client-only (serverless) architecture in which a designated client acts as the student each round, distilling with its own local data, with teachers weighted by a reliability score based on entropy minimization and diversity maximization. On CIFAR-10 under severe label skew, the serverless variants reach 70.28% test accuracy against 43.57% for FedFTG and 43.26% for FedSAM, and adding noise inputs gives no clear further benefit. Despite these gains, training exhibits persistent oscillations; we analyze this convergence behaviour and suggest remedies. Our implementation is available at https://github.com/muhammad-ahmad-1/ATML_G4/tree/Project.

1 Introduction

In classical federated learning (FL), models are trained collaboratively across clients by aggregating their local parameter updates, typically using Federated Averaging (FedAvg) [McMahan et al., 2017], allowing training without data transfer. However, when client data is non-independent and identically distributed (non-IID), local updates may conflict, leading to a divergence from the global optimum. FedAvg relies on simple parameter averaging, which overlooks the complexities introduced by mismatched data distributions and often produces global models that fail to generalize across clients. The permutation invariance of neural network architectures and the diverse learning trajectories of client models further complicate aggregation, as models may update parameters with similar outputs in incompatible ways, contributing to client drift.

We therefore investigate an online, multi-teacher, single-student distillation framework that rectifies (or replaces) parametric averaging after each training round. Local client models act as teachers and the global model as the student; knowledge is distilled through logit matching, which explicitly enforces the global model to match the common input–output relationships of the clients. Our starting hypothesis was that this distillation could be made entirely *data-free* by using noise as input [Raikwar and Mishra, 2022], removing the need for the generators or shared synthetic data required by prior federated distillation methods.

The experiments tell a different story. In a standard client–server architecture, where the server has no data, noise-based distillation underperforms all baselines, including FedAvg. The limiting factor is the absence of real data during distillation, not the distillation itself: when a designated client performs the distillation using its own local data, in a decentralized, client-only architecture, the global model improves substantially and noise becomes unnecessary. Our contributions are:

- We extend data-free, noise-engineered knowledge distillation (with Gaussian and structured noise) to FL as a replacement for, or rectification of, parametric aggregation, and show that in a client–server setting it is insufficient on its own, underperforming FedAvg, FedSAM and FedFTG.
- We propose a decentralized (serverless) architecture in which a designated client, selected cyclically or at random, distills the client models using its local data. In our setting (CIFAR-10, Dir(0.1), 10 clients) it reaches 70.28% test accuracy against 34.91–43.57% for the baselines.
- We adapt entropy-minimization and diversity-maximization objectives from source-free domain adaptation into a reliability score for weighting teachers. Its effect was marginal in centralized tests and not conclusive in the federated runs.
- We analyze the convergence behaviour of the serverless approach, tracing its oscillations to the balance between the distillation and cross-entropy terms, and outline remedies.

2 Related Work

Heterogeneity in federated learning. Recent work on data heterogeneity explicitly regulates the deviation of local models from the global model [Li et al., 2020], uses control variates [Karimireddy et al., 2020], harmonizes updates before aggregation to overcome conflicts [Zhang et al., 2024], or seeks flat minima through stochastic weight averaging [Caldarola et al., 2022] and sharpness-aware minimization [Qu et al., 2022, Fan et al., 2024]. Each method retains parametric averaging to some extent. An attractive alternative is to remedy the aggregation itself by focusing on the models’ functional mapping, i.e., their logits, since parametric averaging is inherently ill-suited to the permutation invariance of neurons and filters: models can produce identical feature maps or logits despite differing parameters.

Distillation in federated learning. Distillation typically uses an unlabeled proxy dataset on the server, which may not be available. FedGen [Zhu et al., 2021] trains a lightweight server-side generator to synthesize global data representations that are shared to guide local training, at additional overhead. FedDistill [Song et al., 2024] distills the global model into local models with a group distillation objective that focuses on under-represented classes, but degrades with small local datasets. FedFTG [Zhang et al., 2022], closest in spirit to our approach, fine-tunes the global model on the server by distillation, using a generator trained to produce hard samples on which the global and local models disagree; it again requires an additional generator.

Data-free knowledge distillation. In centralized settings, Zero-Shot Knowledge Distillation [Nayak et al., 2019] synthesizes “Data Impressions” from the teacher through iterative backpropagation, and Braun et al. [2024] obtain samples in a similar way, but producing these impressions is computationally expensive. Noise-Engineered Knowledge Distillation (NEKD) [Raikwar and Mishra, 2022] instead uses artificial noise as a substitute for training data, which suits settings such as FL where the original data is unavailable. It is, however, limited to models with BatchNorm layers and requires large batch sizes to compute the batch statistics that align the noise with the training distribution.

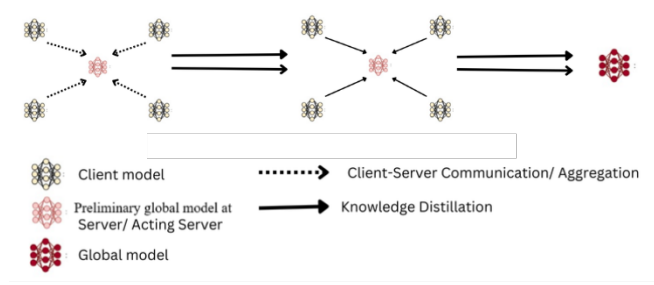


Figure 1: Overview. Client models are aggregated into a preliminary global model at the server (or at the acting server in the serverless variant), which is then refined by multi-teacher knowledge distillation from the client models.

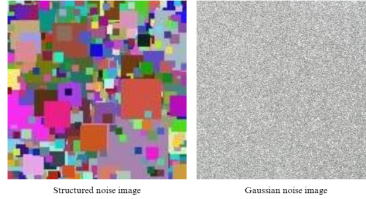


Figure 2: Structured (Dead Leaves–Squares) and Gaussian noise inputs.

3 Method

Figure 1 summarizes the framework. In each round, clients train locally and send their models to an aggregating node, which forms a preliminary global model and then refines it by distilling from the client models.

3.1 Distillation as aggregation

All clients are initialized with the same model and train on their local data for E epochs. The client models $\{T_i\}$ are then used as teachers to distill into a student S , which is either the current round’s parametric aggregate (**Rectify**) or the previous round’s global model (**Replace**). The student is updated by minimizing the (weighted) knowledge distillation loss $\mathcal{L}_{KD}$ between teacher and student logits over the distillation inputs, and the updated student is broadcast to all clients. Over multiple rounds, this repeatedly rectifies the global model by enforcing output matching rather than parameter matching.

3.2 Noise-engineered distillation inputs

NEKD [Raikwar and Mishra, 2022] uses Gaussian noise as input. BatchNorm layers normally normalize with running statistics computed on real data, so feeding noise induces a covariate shift in hidden activations. NEKD replaces the running statistics with batch statistics computed on the noisy inputs, so that the recalibrated BatchNorm layers align hidden activations with those expected on real data and the propagated activations convey meaningful information. Raikwar and Mishra [2022] also show that structured noise gives marginally better results than pure noise; following them, we additionally use the Dead Leaves–Squares dataset [Baradad et al., 2021], in which squares are positioned uniformly at random until the canvas is covered (Figure 2). In the client–server architecture, the server holds no data, so distillation uses noise alone, with uniform teacher weights.

3.3 Serverless architecture

Centralized experiments show that pure-noise distillation is suboptimal compared to distillation with scarce data plus noise [Raikwar and Mishra, 2022], and our federated experiments show that pure-noise methods underperform even FedAvg (Section 5). We therefore adopt a decentralized, serverless

Algorithm 1: Serverless knowledge distillation for FL

Input: Clients $\{C_1, \dots, C_M\}$; local epochs E ; distillation epochs D ; weights $\lambda_1, \lambda_2, \gamma_d, \gamma_n, \beta$; mode $\in \{\text{Rectify, Replace}\}$

```
1 for communication round  $r = 1, \dots, R$  do
2   Select designated client  $C_d$  (cyclically or randomly);
3   Train all (or a random subset of) clients for  $E$  epochs; send teachers  $\{T_i\}$  to  $C_d$ ;
4    $S \leftarrow$  parametric average of  $\{T_i\}$  if Rectify else previous global model;
5   for  $d = 1, \dots, D$  do
6     for each local batch  $(x, y)$  of  $C_d$  (and noise batch  $g$ ) do
7       Compute reliability weights  $R(T_i)$  on  $x$  (Eqs. 2–3);
8       Update  $\theta_S$  on Eq. (1);
9   Broadcast  $S$  to all clients;
```

scenario: there is no central server; instead, a designated client receives the models of all (or a subset of) other clients, aggregates them, and performs the distillation locally. To avoid centralization on fixed participants, the role rotates cyclically when there are few clients, and randomly in larger setups. The designated client can use its own higher-quality local data for distillation, optionally supplemented with noise, together with a cross-entropy term on its labels. The objective for the student parameters θ_S is

$$\min_{\theta_S} \gamma_d \sum_i R(T_i) \mathcal{L}_{KD}(T_i(x), S(x)) + \gamma_n \frac{1}{M} \sum_i \mathcal{L}_{KD}(T_i(g), S(g)) + \beta \mathcal{L}_{CE}(y, S(x)), \quad (1)$$

where (x, y) are local batches of the designated client, g are noise inputs, M is the number of teachers, $R(T_i)$ are the teacher reliability weights (Section 3.4), and $\gamma_d, \gamma_n, \beta$ weight the data-distillation, noise-distillation, and cross-entropy terms ($\gamma_n = 0$ for data-only distillation). Algorithm 1 gives the full procedure.

3.4 Reliability-weighted teachers

Uniform teacher weighting assumes each local model is equally good; under heterogeneity, some clients learn better mappings, and different local distributions produce inconsistent outputs for the same inputs, so uniform weighting may face the same issues as parametric averaging. A reliable teacher should be confident (a peaked output distribution), but not trivially so: a model trained on a single class may predict that class for every input. A similar problem arises in source-free unsupervised domain adaptation (SFUDA) [Hou and Zheng, 2021], which uses an entropy-minimization loss [Grandvalet and Bengio, 2004] to encourage confident predictions and a diversity-enforcing loss [Roy et al., 2022, Wu et al., 2020] to prevent collapse onto the most frequent categories. We use both as a proxy for teacher reliability. For a batch of N inputs with teacher probabilities $p_{i,c}$ over C classes,

$$\mathcal{L}_{ent} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C p_{i,c} \log p_{i,c}, \quad \mathcal{L}_{div} = -\frac{1}{C} \sum_{c=1}^C \hat{q}_c \log \hat{q}_c, \quad \hat{q}_c = \frac{1}{N} \sum_{i=1}^N p_{i,c}, \quad (2)$$

and $\mathcal{L}_{rel} = \lambda_1 \mathcal{L}_{ent} - \lambda_2 \mathcal{L}_{div}$, where λ_1 and λ_2 control the relative importance of confidence and diversity. The reliability of teacher i is the inverse of this loss, normalized over teachers and passed through a softmax:

$$r_i = (\mathcal{L}_{rel}^i)^{-1}, \quad R(T_i) = \text{softmax}_i(r_i / \sum_j r_j). \quad (3)$$

Noise inputs yield diverse, relatively uniform logits even after the covariate shift is reduced, so adaptive weighting is applied only to data-based distillation; noise-based distillation uses uniform weights.

4 Experimental Setup

Data and model. We use CIFAR-10 with a Dirichlet split $\text{Dir}(\alpha)$ over label ratios, where smaller α means higher heterogeneity; all experiments use the highly heterogeneous $\alpha = 0.1$. The backbone

Table 1: Hyperparameters.

Clients / participation / rounds	$K = 10$ / full ($C = 1$) / $T = 15$
Local training	$E = 10$ epochs, batch size 100, learning rate 10^{-3} , weight decay 10^{-3} ; global learning rate 1
Distillation	$D = 10$ epochs (unless stated), Adam, learning rate 10^{-3} , batch size 256
Loss weights (Eq. 1)	$\beta = 0.2$ (initially 3; Section 5.4), $\gamma_d = 0.8$, $\gamma_n = 0.3$
Reliability weights	$(\lambda_1, \lambda_2) \in \{(1, 0), (0.7, 0.3), (0.3, 0.7)\}$
Designated client	cyclic (random selection in ablations)

Table 2: Best top-1 test accuracy (%) of the final global model on CIFAR-10, Dir(0.1). Client–server rows use structured noise with Rectify; serverless rows distill on local data alone with Rectify and cyclic selection.

Algorithm	Accuracy
<i>Centralized learning</i>	87.67
FedAvg	34.91
FedSAM	43.26
FedFTG	43.57
NEKD-FedAvg, client–server	26.01
NEKD-FedSAM, client–server	33.32
NEKD-FedAvg, serverless	70.28
NEKD-FedSAM, serverless	69.65

is a reduced ResNet-18 with one block per layer (instead of two) and filter widths [32, 32, 64, 64] (instead of [64, 128, 256, 512]).

Baselines. We compare the client–server and serverless variants with FedAvg [McMahan et al., 2017], FedSAM [Qu et al., 2022], and FedFTG [Zhang et al., 2022], the existing federated distillation method closest to ours in motivation, and report centralized training as an upper bound. Each distillation variant is run with SGD (“FedAvg”) or SAM (“FedSAM”) local training, to test orthogonality to client-side optimizers.

Hyperparameters and metrics. Settings are held constant across experiments (Table 1). We report the accuracy and F1 score of the global model on test data after each round, and the variance of gradients (the average distance between local and global updates after distillation). During distillation we also track the individual loss terms and the alignment of the global model with the client models.

5 Results

5.1 Data-free distillation is insufficient; local data is the key factor

Table 2 reports test accuracy for all methods. The client–server variants, which distill from noise alone, underperform all baselines, including FedAvg; this holds for both noise types and for both Replace and Rectify (21.34–35.89%; Table 3). While they improve slowly and do not plateau at a lower accuracy, noise-engineered distillation techniques from centralized learning do not by themselves suffice to replace parametric aggregation. In complete contrast, the serverless variants significantly outperform all baselines (70.28% and 69.65%), showing the importance of real data during distillation.

Within the serverless architecture, the differences between noise types and between Replace and Rectify (Table 3) show no clear trend and fall within the range of the round-to-round oscillations discussed in Section 5.4; the significant factor is the use of client-side data. Random selection of the designated client gives results as competitive as cyclic selection, confirming that the gains are not an artifact of systematic selection (Appendix B). Noise does not improve final accuracy, but it does increase the oscillations in performance. In small-scale settings, local data alone is therefore sufficient for distillation; whether larger setups require supplementing it with noise remains to be tested.

Table 3: Test accuracy (%) across noise types and aggregation modes. These are separate runs from Table 2; serverless results were obtained after the β fix. [†]Obtained with 20 distillation epochs; all other entries use 10.

Architecture	Algorithm	Gaussian noise		Structured noise	
		Replace	Rectify	Replace	Rectify
Client-server	NEKD-FedAvg	21.62	21.34	23.68	23.95
	NEKD-FedSAM	30.83	32.27	34.41	35.89
Serverless	NEKD-FedAvg	66.69	61.32	75.01 [†]	68.81
	NEKD-FedSAM	60.02	65.33	68.27	69.78

Table 4: Effect of distillation epochs (serverless, data + structured noise, cyclic selection).

Distillation epochs	1	2	5	10	20
Accuracy (%)	46.42	54.39	59.56	68.44	70.31

5.2 Effect of distillation epochs

Apart from the architecture, the number of distillation epochs is the most critical choice (Table 4). Too few epochs (e.g., 1) limit improvement, as the global model has insufficient exposure to the client models, while too many (e.g., 20) lead to convergence issues despite competitive accuracy. Longer distillation also increases the variance of updates, pulling the global model further from the client models in parameter space while aligning their outputs (Appendix B). This is consistent with our aim of functional rather than parametric consistency.

5.3 Reliability weighting

The effect of adaptive teacher weighting could only be explored in a limited fashion. The configurations tried in the federated runs gave similarly high performance and similar convergence issues, so we cannot make strong claims about its effect in data-based distillation. In centralized multi-teacher tests (Appendix D), adaptive weighting performs similarly to uniform weighting, outperforming it marginally when entropy is emphasized (76.37% vs. 75.95% at $\lambda_1 = 0.99$) and in pure-noise distillation.

5.4 Convergence

We initially used $\beta = 3$ for the cross-entropy term, which caused significant performance swings and no correlation between gradient variance and performance gains: the cross-entropy term was over-weighted relative to distillation, so the global model mainly fitted the designated client’s local distribution while forgetting information from previous rounds. Reducing β to 0.2 resolved the wild swings, giving steady improvement and a decrease in gradient variance that correlates with performance gains, although fluctuations persist (Figure 3). Dense logging revealed that while the training loss decreased, the reduction came primarily from the cross-entropy term; the distillation loss increased within rounds and changed randomly across rounds. An explicit drift penalty discouraging deviation from the starting global model did not remove the fluctuations, possibly due to under-weighting (Appendix C). Across configurations, test accuracy trends upward and gradient variance downward, so the method has the desired general direction.

6 Discussion and Future Work

Our initial motivation was a completely data-free distillation setup that uses pure noise to remedy the limitations of parametric averaging. Noise-based methods underperformed in client-server setups, whereas the serverless architecture surpassed all baselines; within it, noise was unnecessary, since local data alone (in sufficient quantity) was enough for distillation and is used without leaving its client. The distillation step has compute similar to one to two additional local training rounds and runs on the client side.

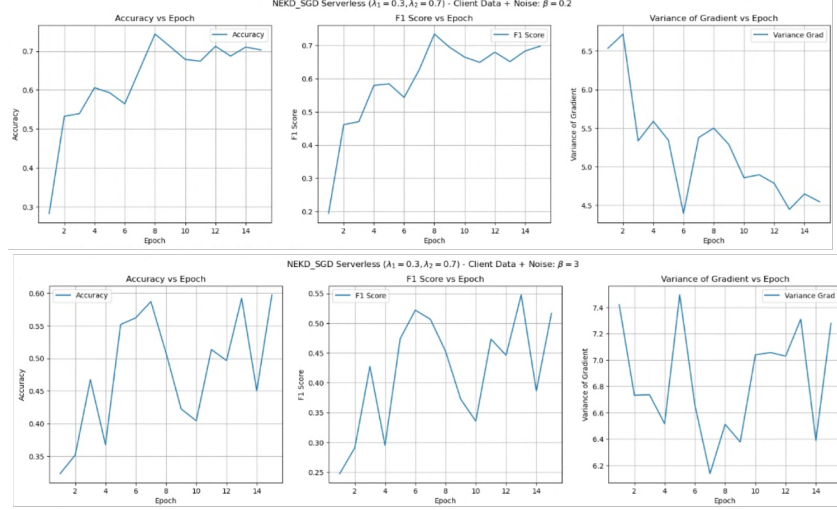


Figure 3: Effect of the cross-entropy weight β on stability: accuracy, F1 score, and gradient variance per round for serverless NEKD with SGD, client data and noise, $(\lambda_1, \lambda_2) = (0.3, 0.7)$, with $\beta = 0.2$ (top) and $\beta = 3$ (bottom).

These results come from a small-scale setting: one dataset, one heterogeneity level, 10 clients, and 15 rounds. Establishing them, and determining whether noise-engineered distillation becomes necessary alongside local data at scale, requires experiments with more clients and rounds, partial participation, and multiple datasets and heterogeneity conditions. Equally important is convergence: hyperparameter tuning mitigated the oscillations only partially, which calls for revisiting the distillation process itself (and how noise enters it), a more thorough search over the reliability and loss weights, and better-tuned drift control.

7 Conclusion

We studied knowledge distillation as a way to rectify or replace parametric aggregation in heterogeneous federated learning. Extending noise-engineered, data-free distillation to FL, we found that it is insufficient on its own in a client-server setting. A serverless architecture in which a designated client distills the client models using its own local data, with reliability-weighted teachers, substantially outperformed FedAvg, FedSAM and FedFTG under severe label skew, with quick gains within a few distillation epochs and without synthetic data generation or additional models. Convergence oscillations remain unresolved; addressing them and scaling to larger setups are needed to confirm the viability of this approach as an alternative to parametric averaging.

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A Noise-Engineered Knowledge Distillation Algorithms

NEKD [Raikwar and Mishra, 2022] aligns noise inputs with the training input distribution using the current batch statistics and the learned BatchNorm scale and shift parameters. Algorithm 2 uses noise alone (client–server variant); Algorithm 3 combines noise with data, as available to the designated client in the serverless variant.

Algorithm 2: NEKD: noise only [Raikwar and Mishra, 2022]

Input: Teacher T , learning rate η

Output: Student $S(\theta)$

```

1 for  $b = 1, \dots, B$  do
2    $g \sim \mathcal{N}(0, 1)$ ; ▷ Gaussian or structured noise
3    $y_T^g \leftarrow T(g \mid \mu_g, \sigma_g)$ ;  $y_S^g \leftarrow S(g \mid \theta, \mu_g, \sigma_g)$ ; ▷ batch statistics
4    $\theta \leftarrow \theta - \eta \partial \mathcal{L}_{KD}(y_T^g, y_S^g) / \partial \theta$ ;

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Algorithm 3: NEKD: noise and data [Raikwar and Mishra, 2022]

Input: Teacher T , weights $\gamma_n, \gamma_d, \beta$, learning rate η

Output: Student $S(\theta)$

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1 for  $b = 1, \dots, B$  do
2    $g \sim \mathcal{N}(0, 1)$ ;  $(x, y) \leftarrow \text{local batch}$ ;
3    $\theta \leftarrow \theta - \gamma_n \eta \partial \mathcal{L}_{KD}(T(g), S(g)) / \partial \theta$ ; ▷ noise KD
4    $\theta \leftarrow \theta - \gamma_d \eta \partial \mathcal{L}_{KD}(T(x), S(x)) / \partial \theta - \beta \eta \partial \mathcal{L}_{CE}(S(x), y) / \partial \theta$ ; ▷ data KD + CE

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We also replicated Zero-Shot Knowledge Distillation [Nayak et al., 2019], which crafts Data Impressions by sampling softmax vectors from a Dirichlet distribution parameterized by the teacher’s class-similarity matrix and optimizing random-noise images to match them, in centralized learning. We did not extend it to FL because of the careful tuning and extensive computation required to produce quality Data Impressions.

B Detailed Ablations

Client–server training curves. Figure 4 shows the client–server variants with Rectify and Replace: accuracy improves slowly but steadily, and gradient variance decreases, similar to FedSAM.

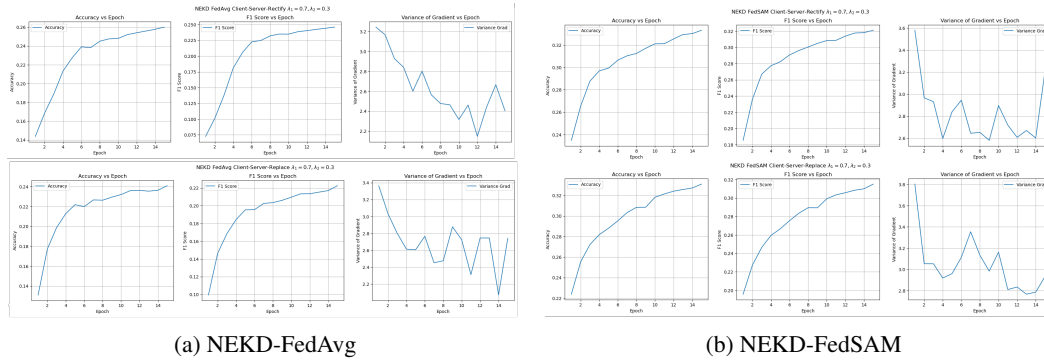


Figure 4: Client–server variants with Rectify (top) and Replace (bottom), $(\lambda_1, \lambda_2) = (0.7, 0.3)$.

Client selection. Random selection of the designated client gives similar, if not better, results than cyclic selection (Figure 5).

Distillation epochs. Figure 6 shows how test accuracy and the variance of updates change with the number of distillation epochs. While performance increases with more epochs, so does the

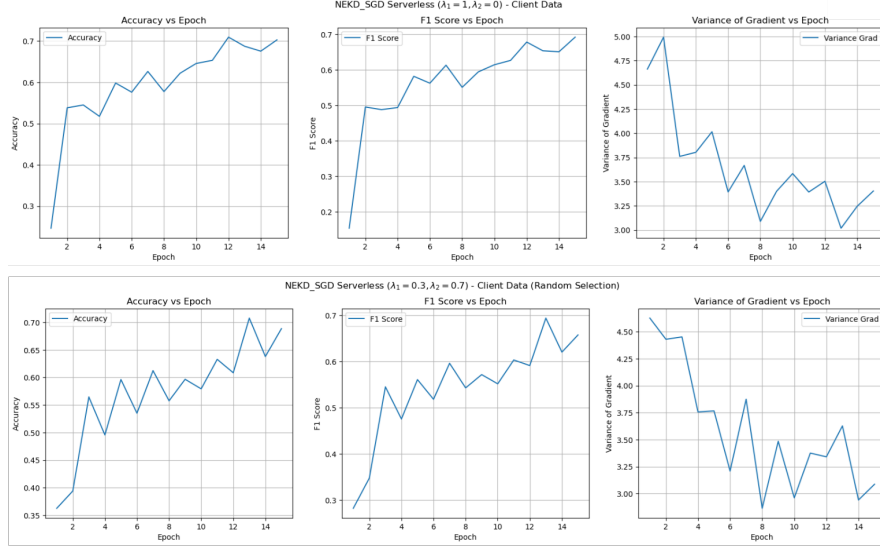


Figure 5: Serverless NEKD-FedAvg with cyclic selection, $(\lambda_1, \lambda_2) = (1, 0)$ (top), and random selection, $(0.3, 0.7)$ (bottom).

magnitude of update variance: longer distillation pulls the global model further from the distilling client models in parameter space while aligning their outputs. As our goal is functional rather than parametric consistency, a decreasing trend in variance is what indicates convergence. Consistent with the limitations of parametric aggregation, the client–server distillation variant (lowest accuracy) has the lowest variance magnitude of our implementations.

C Convergence Analysis

Gradient metrics of the baselines. The variance of FedAvg increases over rounds while that of FedSAM decreases (Figure 7): FedSAM mitigates client drift, whereas FedAvg suffers worse drift over rounds. The trend of this metric can indicate convergence toward a better minimum.

Serverless variants with noise and drift penalty. Figures 8a–8b show three serverless configurations, with SGD and SAM local training: distillation with data alone, data and noise, and data, noise and a drift penalty (a regularization term discouraging drift from the starting global model). Noise leads to substantially more prominent fluctuations, particularly with SAM, and the drift penalty did not remove them. Across all configurations, test accuracy increases and gradient variance decreases overall.

D Centralized Pilot Experiments

Uniform vs. adaptive weighting. In a multi-teacher, single-student setup with five teachers distilling into a ResNet-18 on CIFAR-10 (Table 5), adaptive weighting performs similarly to uniform weighting regardless of (λ_1, λ_2) , and marginally outperforms it for high λ_1 : emphasizing entropy, i.e., teacher confidence, matters more than diversity. We also tested pure noise and noise with 10% of the data, adapting the NEKD implementation of Raikwar and Mishra [2022]. Adaptive weighting marginally outperforms uniform weighting throughout with pure noise; with data added, the results are inconclusive and suggest the need for a hyperparameter search, which may be dataset-dependent.

Data-based vs. noise-based distillation. In a one-teacher, one-student setting on CIFAR-10 (pre-trained ResNet-32 teacher, ResNet-18 student, learning rate 10^{-3} , 20 epochs), data-based distillation outperforms noise-based distillation (NEKD) as expected (Table 6). The accuracy of noise-based distillation, however, rises much faster: by 28 points between epochs 5 and 20, versus 6 points for data-based distillation. This suggested that more epochs could bring it closer to data-based

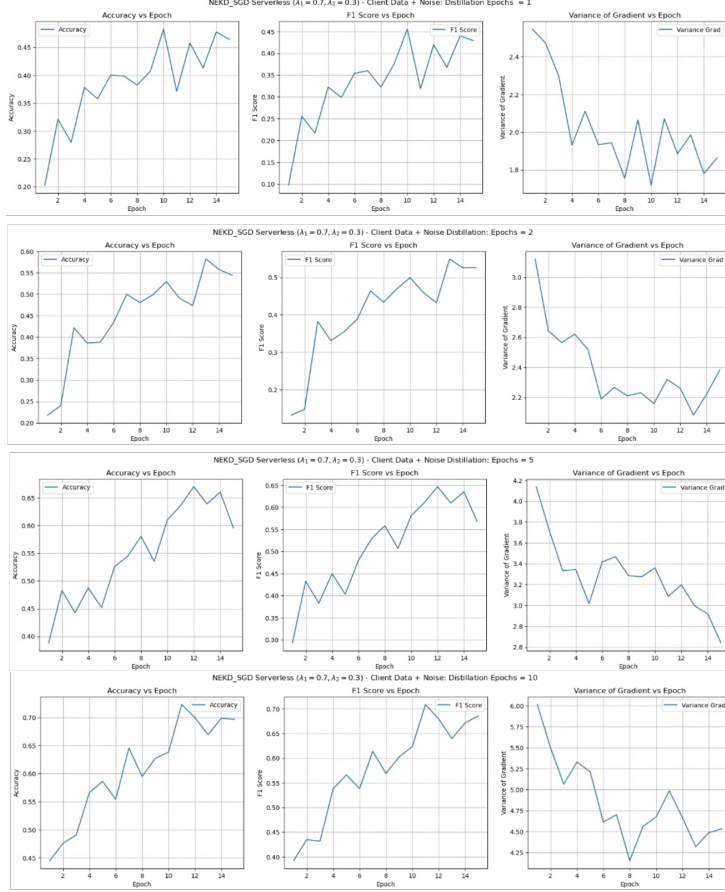


Figure 6: Serverless NEKD-FedAvg (client data + noise, $(\lambda_1, \lambda_2) = (0.7, 0.3)$) with 1, 2, 5, and 10 distillation epochs (top to bottom).

Table 5: Centralized multi-teacher distillation: test accuracy (%) by weighting scheme.

Weighting	(λ_1, λ_2)	Data	Noise	Noise + 10% data
Uniform	—	75.95	13.26	47.76
Adaptive	(0.7, 0.3)	76.09	14.95	46.99
Adaptive	(0.5, 0.5)	74.39	15.31	50.73
Adaptive	(0.3, 0.7)	67.24	14.02	51.35
Adaptive	(0.01, 0.99)	73.94	14.76	27.89
Adaptive	(0.99, 0.01)	76.37	13.73	34.42

performance, and that even a small portion of real data could lift data-free distillation in the federated setting.

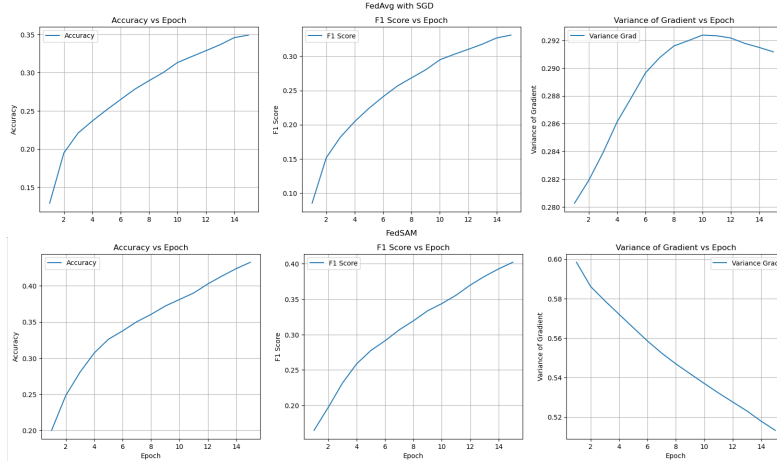


Figure 7: FedAvg (top) and FedSAM (bottom): accuracy, F1 score, and gradient variance per round.



Figure 8: Serverless NEKD with data only (top), data + noise (middle), and data + noise + drift penalty (bottom).

Table 6: Centralized one-teacher distillation on CIFAR-10: validation accuracy (%) and losses.

Epoch	Original training data			Gaussian noise (NEKD)		
	Train loss	Val. loss	Val. acc.	Train loss	Val. loss	Val. acc.
5	0.4965	0.6853	77.07	2.2134	1.8855	33.88
10	0.2274	0.5885	81.91	2.1236	1.6669	42.46
15	0.1861	0.5853	81.96	2.0345	1.3326	55.76
20	0.1705	0.5493	82.91	1.9624	1.1579	62.50